

Smart Multi-Maintenance & Multi-Connect Ecosystem

SMART-2MCE

"Connecting Maintenance. Empowering Ecosystems."

700+

Institutions Targeted

30

Districts Covered

3

Access Channels

50,000+

Year-1 Beneficiaries

TEAM LEAD

INSTITUTION

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LIVE DEMO: smart-2mce.vercel.app | Prototype Already Running

Submitted to: DSE Innovation Hackathon 2026 ·
SOLVIT AFRICA

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00 BASIC INFORMATION

Project Name	SMART-2MCE — Smart Multi-Maintenance & Multi-Connect Ecosystem
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Category	Technology / Digital Services / Social Innovation

01 PROBLEM STATEMENT

Over 700 public institutions in Rwanda — hospitals, universities, government offices, and NGOs across all 30 districts — lose millions of RWF annually because **no structured digital system exists** for maintenance management, asset tracking, or technician deployment. Simultaneously, **thousands of qualified ICT, electrical, and mechanical graduates remain unemployed** despite real maintenance demand existing in every district.

No Asset Visibility	ALL SECTORS	Devices and materials in 700+ institutions are completely unregistered. Nobody knows what exists, where it is, or its condition — leading to loss, duplication, and zero lifecycle tracking.
Maintenance Delays	PUBLIC & NGO	Repairs wait days or weeks with no accountability. Hospitals operate broken ICU equipment. Universities run classes without functioning projectors. No deadline. No tracking. No urgency.
Zero Accountability	ALL SECTORS	No audit trail exists. Nobody knows who repaired what, when, or at what cost — creating direct opportunities for fraud and resource waste in public institutions.
Graduate Unemployment	INDIVIDUALS	Thousands of ICT, electrical, and mechanical graduates across Rwanda cannot access maintenance work despite real demand — because no structured marketplace connects them to institutions.

02 PROPOSED SOLUTION

Overpriced Procurement	ALL SECTORS	No competitive supplier market exists. Institutions consistently overpay for spare parts and materials due to complete lack of price transparency and structured bidding.
Broken Communication	ALL SECTORS	WhatsApp messages and phone calls between institutions and technicians cause disputes, lost agreements, and zero legal traceability of maintenance commitments.
Rural Digital Exclusion	RURAL AREAS	Existing digital solutions exclude technicians and staff in low-connectivity areas who lack smartphones — meaning the people who need economic opportunity most are locked out.

SMART-2MCE is a national-scale digital platform that unifies Institutions, Technicians, and Suppliers inside one intelligent, transparent service ecosystem — accessible via **Web, Mobile App, and USSD** so that every Rwandan, regardless of device or connectivity, can participate fully.

Step 1	Institution registers and adds all devices with unique ID codes (e.g. KRH/MH/001) and materials by room — creating a complete national asset registry.		
Step 2	Institution creates a maintenance request with title, linked device, problem description, required skills, deadline, priority level (LOW / MEDIUM / HIGH / CRITICAL), and budget range.		
Step 3	Platform checks if an internal technician is available. If yes, task is assigned directly. If not, task is published to the external marketplace.		
Step 4	Unemployed graduate technicians browse available tasks filtered by district and skill type — via Web, Mobile App, or USSD on any phone — and submit proposals with price and timeline.		
Step 5	Institution reviews all proposals side by side and opens a negotiation thread with the preferred candidate. Terms are agreed and technician is formally accepted.		
Step 6	If spare parts are needed, technician raises a parts request. Suppliers receive alerts and submit competitive quotes. Institution selects the best quote and confirms delivery.		
Step 7	Technician completes work and marks it done. Institution verifies quality. Both parties rate each other. Every action is permanently logged in the immutable audit trail.		
WEB PLATFORM		MOBILE APP	USSD *844#
Full dashboards, analytics, marketplace, AI assistant — for institution admins and desktop users.		Real-time alerts, task management, marketplace proposals — on any Android or iOS smartphone.	Zero internet needed. Report faults, check task status, confirm jobs — on any basic feature phone in rural Rwanda.

03 KEY FEATURES (Max 5)

02 FEATURE	11-Stage Intelligent Task Lifecycle with Countdown Timers WHAT IT DOES: Every maintenance request moves through 11 governed stages — from CREATED to CLOSED — with automatic countdown timers, multi-level deadline alerts at 50%, 80%, 95%, and 100%, and automatic LATE flagging. WHY IT MATTERS: Delays become impossible to hide. Institutions always know exactly what is happening with every repair. Accountability is enforced by the system — not by human memory.
01 FEATURE	USSD Zero-Internet Inclusion (*844#)

WHAT IT DOES: Any user dials *844# to report a fault, check task status, submit a proposal, or confirm a job — with no smartphone or internet connection required.

WHY IT MATTERS: Rwanda has millions of feature phone users in rural districts. This single feature ensures that the most economically vulnerable graduates and institution staff are never excluded from the ecosystem — making SMART-2MCE the only maintenance platform with true national reach.

03

FEATURE

Graduate Employment Marketplace — No Auto-Assignment

WHAT IT DOES: Unemployed graduates browse tasks filtered by district and skill, submit proposals with price and timeline, negotiate terms through task-scoped chat, and get formally hired after institution approval.

WHY IT MATTERS: Directly converts Rwanda's graduate unemployment crisis into a structured, merit-based income pathway. No algorithm forces matches — technicians choose their work, institutions choose their technicians, preserving fairness and trust.

04

FEATURE

Competitive Supplier Bidding Engine

WHAT IT DOES: When spare parts are needed, all matched suppliers receive automatic alerts and submit competitive quotes ranked by price, delivery speed, and rating. Institution selects the best option.

WHY IT MATTERS: Eliminates procurement overpricing permanently. Institutions save millions of RWF annually that were previously lost to non-competitive, relationship-based purchasing decisions.

05

FEATURE

AI Predictive Maintenance Engine

WHAT IT DOES: Natural-language AI assistant answers institutional queries, detects devices that have failed repeatedly, predicts failures before they occur, recommends the nearest top-rated technician, and generates national performance dashboards.

WHY IT MATTERS: Shifts institutions from reactive repair to proactive maintenance — saving money, extending asset lifespan, reducing downtime, and providing government with real-time national infrastructure health data.

04 VALUE PROPOSITION

No platform in Rwanda or East Africa today combines maintenance management, graduate employment, competitive procurement, AI analytics, and universal Web + Mobile + USSD access in one ecosystem.

SMART-2MCE is not an improvement on existing solutions — it is an entirely new category.

USSD Inclusion — True Universal Access

The only maintenance platform accessible to every Rwandan regardless of phone type or internet connection. A graduate in rural Nyamagabe with a basic feature phone has equal marketplace access to a technician in Kigali with a smartphone. No competitor offers this.

No Auto-Assignment — Human-Centred Marketplace

Unlike generic job platforms, SMART-2MCE never forces algorithmic matches. Technicians choose their work. Institutions choose their technicians. This preserves dignity, trust, and marketplace integrity — critical for formal institutional adoption.

Triple Problem Solver — One Platform

Most solutions solve one problem. SMART-2MCE simultaneously solves maintenance delays, graduate unemployment, overpriced procurement, zero asset visibility, broken communication, and rural digital exclusion — in one integrated, self-reinforcing ecosystem.

Every new institution brings more tasks. More tasks attract more technicians. More technicians attract more institutions. More procurement requests attract more suppliers. The platform becomes more valuable to every actor automatically — without additional marketing cost.

Network Effect Flywheel — Self-Reinforcing Growth		
Rwanda-First by Architecture		Built on Rwanda's official Province-District-Sector-Cell-Village location hierarchy mirroring Irembo.rw, with structured device ID codes matching government asset standards — ensuring immediate institutional compatibility and government adoption confidence.

05 BUSINESS MODEL

SMART-2MCE is architected as a self-sustaining digital business — not a donor-dependent project. Five independent revenue streams ensure financial sustainability from Month 4, with no reliance on government subsidies, sponsorships, or promises.

#	REVENUE STREAM	WHO PAYS	HOW MUCH	WHY THEY PAY
1	Institutional Subscription	Hospitals, Universities, Government Offices, NGOs	15,000–80,000 RWF / month (tiered by size)	Saves far more in reduced maintenance costs and eliminated overpricing
2	Task Commission	Technicians on completed tasks	5% per completed task	Platform delivered the job they could not find otherwise

ECONOMIC IMPACT

- Direct job creation for 500+ unemployed graduates across all 30 districts — connecting them to real, paid maintenance work through the technician marketplace.
- Sustainable business growth for 200+ local spare parts suppliers and equipment vendors through competitive bidding and guaranteed institutional demand.
- Institutions save millions of RWF annually through transparent supplier competition and predictive maintenance — eliminating overpricing and premature replacements.
- Formalization of Rwanda's informal maintenance sector — creating verifiable professional profiles and earning histories that unlock financial inclusion for technicians.

3	Supplier Transaction Fee	Suppliers on confirmed orders	3% per fulfilled order	Platform delivered a confirmed paid customer — zero marketing cost
4	Premium Analytics	Government agencies, large hospital networks	200,000 RWF / month	National dashboards save millions in avoided equipment failures
5	USSD Telecom Partnership	MTN Rwanda & Airtel Rwanda	Usage-based billing per session	Drives network usage and digital transformation alignment

BREAK-EVEN PROJECTION		Month 4 Target	Year 1 Target
30 institutions × 50,000 RWF/month		1,500,000 RWF / month	Platform covers all operational costs
100 tasks × 5% commission at avg 80,000 RWF		400,000 RWF / month	500+ active technicians earning income

06 SOCIAL IMPACT — REAL NUMBERS

500+	700+	50,000+	500,000+
Graduate Technicians Earning Income (Year 1)	Institutions with Full Asset Visibility (Year 1)	Direct Beneficiaries (Year 1)	Beneficiaries by Year 3 (National)

Web Platform (Live)
✓ Institution Admin Dashboard with KPI cards, live countdown timers, and late-task alerts
✓ Device Registry with ID-based tracking (e.g. UR/CS/001 — Dell Laptop — CS Lab 1 — Working)
✓ Service Request creation form with priority levels, deadline, linked device, and budget
✓ Technician Marketplace with task listings, skill filters, district filters, and Apply button
✓ Supplier Dashboard with procurement requests, quote submission, and order tracking
✓ Authentication with JWT + OTP two-factor verification and role-based access control
Mobile App (In Development)

✓ Real-time push notifications for new tasks, accepted proposals, and deadline alerts

• Revenue generation through platform subscriptions and commissions — building a self-sustaining Rwandan technology business with continental expansion potential.

SOCIAL IMPACT

• Improved learning, research, and healthcare environments through faster maintenance of critical infrastructure — directly benefiting students, patients, lecturers, and staff.

• Complete digital inclusion through USSD — ensuring technicians and institution staff in all 30 districts participate equally regardless of smartphone ownership.

• Merit-based marketplace gives new graduates equal opportunity to build professional reputation and income based on skills and performance, not connections.

• Full government transparency — every maintenance action on public assets is permanently logged, timestamped, and reportable to oversight agencies.

• Empowerment of women technicians and rural entrepreneurs through equal, merit-based marketplace access — promoting gender inclusion and geographic equity.

ENVIRONMENTAL IMPACT

• Significant reduction in premature device disposal through predictive maintenance — extending asset lifespan and reducing electronic waste across institutions.

• Repair-first culture replaces replace-first behavior — reducing resource consumption, manufacturing demand, and carbon footprint associated with new equipment production.

• Contribution to Rwanda's national climate commitments by formalizing preventive maintenance as standard institutional practice across all sectors.

• Reduced transport emissions through AI proximity matching — connecting institutions to the nearest qualified technician and fastest local supplier.

07 MVP — MINIMUM VIABLE PRODUCT

**LIVE DEMO — Prototype Already
Running:**

smart-2mce.vercel.app

SMART-2MCE is not just an idea — it is a running system. The full technical ecosystem has been architected across 24 modules with complete database schema (14 tables), 11-stage task lifecycle, role-based dashboards, business rules, API structure, and three-channel access strategy. The prototype is live, interactive, and ready to demonstrate to judges.

✓ Task management — view active assignments, update progress stages, request spare parts

✓ Marketplace browser — find tasks by skill type, district, urgency, and budget

✓ Earnings tracker — income per task, monthly summary, and payment history

USSD Application *844# (Architected)

✓ Menu 1: Report a fault — select institution, describe problem, confirm submission

✓ Menu 2: Check task status — enter task ID, receive current stage and countdown

✓ Menu 3: Find work — receive nearest available tasks matching your skill type

✓ Menu 4: Confirm job acceptance — enter task ID, confirm availability, receive assignment

✓ Works on every phone — MTN, Airtel — no smartphone or internet required

SMART-2MCE does not need permanent funding to survive. It needs funding to ignite.

Once the ecosystem reaches critical mass — 50 institutions, 200 technicians, 100 suppliers — the platform becomes permanently self-sustaining through the compounding value it delivers to every actor within it.

This is not a university project. This is Rwanda's national maintenance infrastructure — built by a Rwandan student, for Rwanda, designed from day one to scale across Africa.

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